



SpiderRock V7 Tool Concept Guide

Chart Tool

July 23, 2018

OVERVIEW

The Chart Tool is designed to assist the user with visually displaying historical trade volume, price and volatility metrics for a symbol or a pair of symbols. The user has access to 10 years of implied and realized volatilities, high / low / open / close prices, trade volume, earnings and more.



LAUNCH SAVED LAYOUT

SAVE CURRENT LAYOUT AS DEFAULT

CHART TOOL

- Visualize historical price, volume, and implied volatility levels
- Earnings history as well as censored and uncensored realized and implied volatilities
- Most data series include ten years or more of history

SYSTEM INFORMATION

User: annabelle (SRAdmin)
Version: v7.0
Citrix Host: SPIDERNJ134
Tool Server: 10.37.201.109:5240
Environment: V7_Stable.Prod | NY4
Time: 04:41:06.0 PM



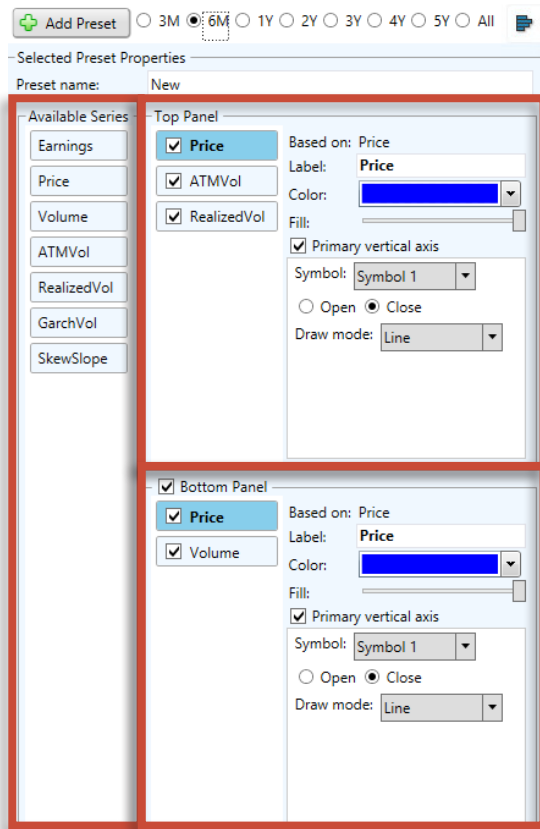
CHART TOOL VIEW



Settings Panel

- This panel allows users to select one or two symbols (Symbol 1 and Symbol 2) to populate the graphs. The first symbol is the default symbol for each timeseries, and the second symbol can be used as a reference or comparison symbol to the first.
- **Load:** loads previously saved Chart Tool settings, which store user-defined presets
- **Save / Save As:** Saves the current settings over the previously loaded Chart Tool settings / Saves the current settings under a new Chart Tool settings entry.
- **Refresh:** refreshes the timeseries data and display (tool does not update data automatically)
- **Add Preset button:** allows users to create a new chart tab that can be named under Selected Preset Properties in the Metrics Panel.
- **Menu Icon button:** clicking this icon will toggle on or off the view of the Selected Preset Properties and Common Properties sidebar.
- **Histogram Toggle button:** clicking this button (below the settings button) displays a histogram representation for the selected series to the right of the Top and Bottom Graphs
- **Timeseries Range:** users can adjust the lookback period in the chart by selecting one of the ranges (3 months - 10 years) displayed to the right of the Add Preset button

CHART TOOL VIEW



Metrics Panel

Available Series

Users can drag and drop metrics from this area to the Top and/or Bottom Panel to create corresponding graphs to the left. In the example displayed here, the user has selected Price, ATMVol, and RealizedVol for the Top Panel and Price and Volume for the Bottom Panel. User can select:

- **Earnings** - displays a vertical bar on each earnings announcement date for the chosen symbol(s)
- **Price** - open, high, low, and close price timeseries
- **Volume** - daily trade volume (Symbol 1 or Symbol 2 only)
- **ATMVol** - SpiderRock Term Structure ATM volatility
- **RealizedVol** - calculated realized volatility for a user-defined rolling window
- **GarchVol** - Garch-style calculated realized volatility for a user-defined rolling window
- **SkewSlope** - the SkewSlope is a calculation of SpiderRock Surface as a rate of change in volatility in terms of moneyness. It is calculated by taking the difference of volatility between the curve points at moneyness -0.5 and +0.5.

Each available series has a set of common characteristics.

- **Label** User-defined label for the chart legend and display
- **Color** User-defined color for the timeseries
- **Primary vertical axis** When this checkbox is activated, the timeseries values will be scaled to the range of values across all timeseries which are assigned to the primary axis. It is useful for visual comparison of similar-type calculations (e.g. realized and implied vols). When the checkbox isn't selected, the values will be scaled independently from others, and the timeseries will have its own unique axis.

The user can configure whether the values reference the first symbol, the second symbol, or a relationship between the two for most available series.

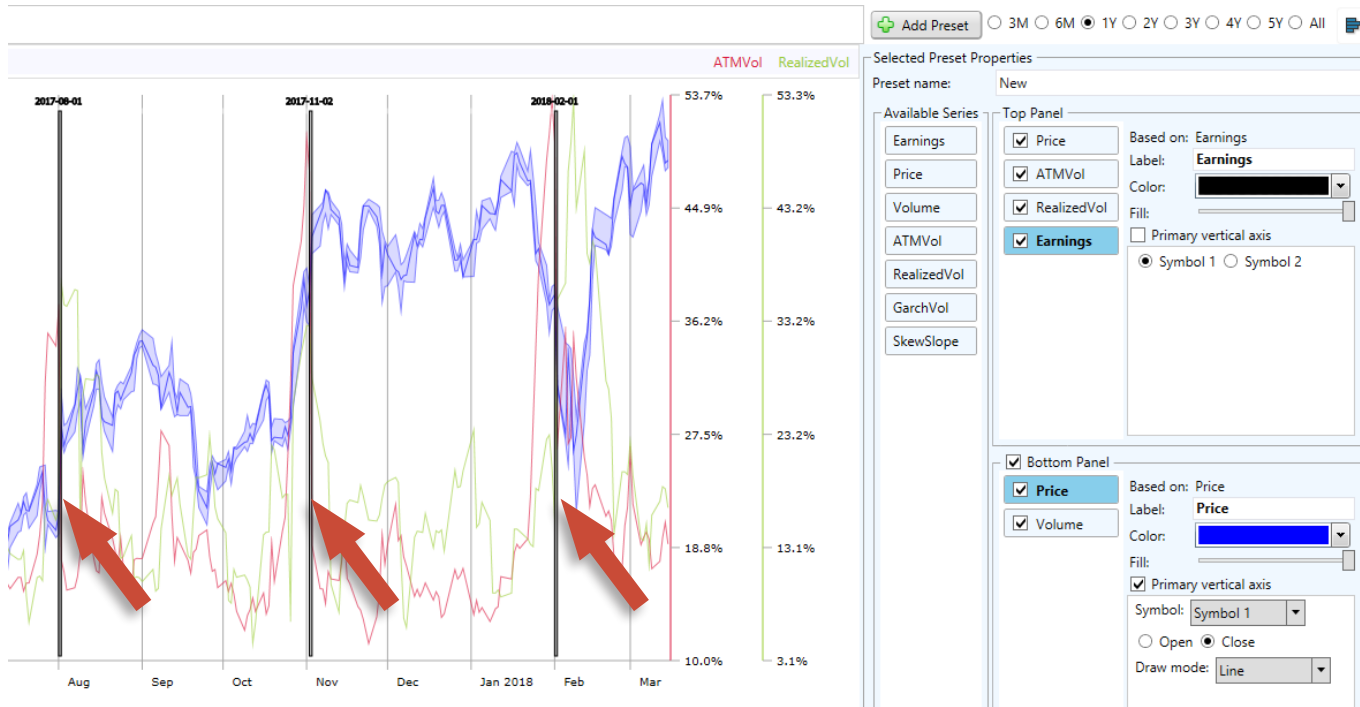
- **Symbol 1** Use the first symbol
- **Symbol 2** Use the second symbol
- **Sym 1 – Sym 2** Use the spread between the first and second symbols
- **Sym 1 / Sym 2** Use the ratio of the first and second symbols
- **Log(Sym 1 / Sym 2)** Use the percent difference between the first and second symbols



AVAILABLE SERIES CONFIGURATION

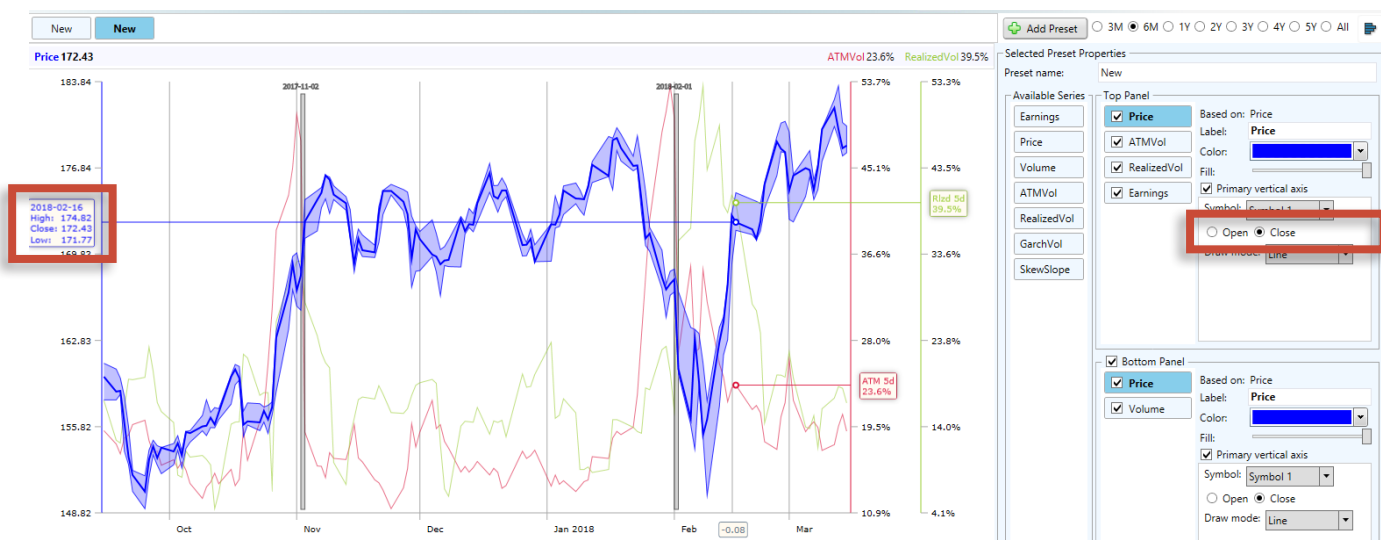
Earnings

Earnings data goes back to 2014 Q2 earnings dates. This series type does not support ratios or any other comparisons between two symbols. Earnings dates (black vertical lines) are shown in the graph below.



Price

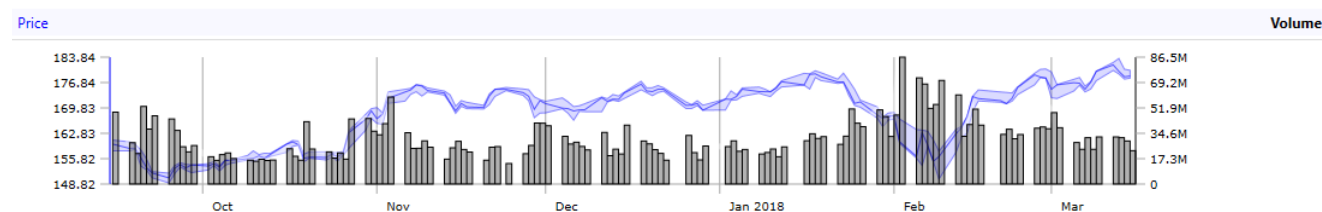
The price timeseries displays open, close, high, and low data for the specific date when a user hovers over a point on the graph. In addition to the common characteristics (color, label, etc), the user can also modify the opacity of the OHLC price overlay, by increasing or decreasing the level of the "Fill" slider. Note that this series type can also be configured to plot relationships between the two symbols.



AVAILABLE SERIES CONFIGURATION

Volume

Trade volume for either Symbol 1 or Symbol 2. This series does not support ratios and other comparisons between the two symbols.



ATMVol

The ATM volatility is measured from the term structure, and values are available from a fixed grid of expirations ranging between 5 trading days to 5 years. By default, the display of ATMVol is uncensored. Users can look at the censored ATMVol (removing Earnings effect) by clicking on the Censored button.



AVAILABLE SERIES CONFIGURATION

RealizedVol

Realized volatility can be calculated for any user-defined trading day window. If the series is configured as a comparison between two symbols, the realized volatility is calculated for each series before evaluating the spread, ratio, or percent.

Clipping

When checked, the full date range of price data is modified so that any daily variance which occurs outside of the user-defined standard deviation multiples is replaced with the variance corresponding to the minimum or maximum standard deviation multiple before calculating the realized volatility.

Censored

When checked, the calculation will ignore earnings moves by removing the daily changes occurring on, before, and after each earnings announcement.

Centered

When checked, the timeseries is shifted left by half the user-defined vol window, so that the realized volatilities are aligned with the middle date of each window.



AVAILABLE SERIES CONFIGURATION

GarchVol

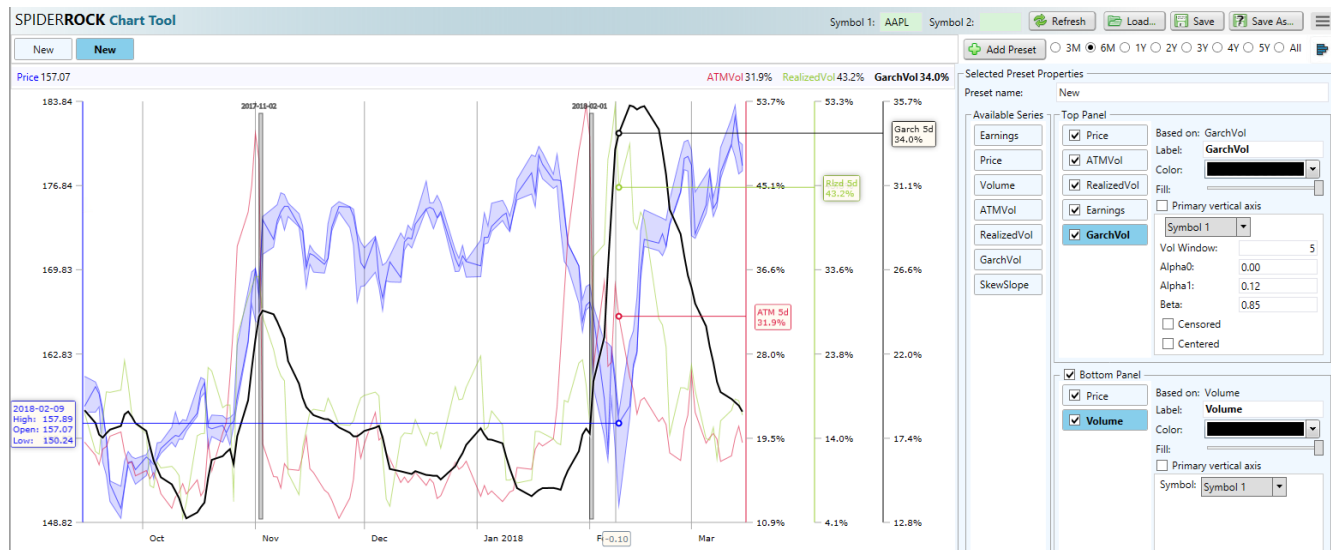
This is an implementation of the GARCH(1,1) model. For an overview of GARCH, see https://en.wikipedia.org/wiki/Autoregressive_conditional_heteroskedasticity#GARCH

The formula for generating daily variance is calculated recursively from the first date in the data range, and the vol window determines the number of variances taken in the average variance used for the calculation of realized vol.

The recursive formula is:

$$\sigma_t^2 = \alpha_0 + \alpha_1 v_{(t-1)}^2 + \beta \sigma_{(t-1)}^2$$

where σ_t^2 is the daily estimate of variance at time t, and $v_{(t-1)}^2$ is the prior day close-close variance. For example, a 5-day vol window would recursively calculate the entire series, and for the realized volatility number, the square root of the average of the 5 most recent observations of σ_t^2 is taken to be the most recent realized volatility number.



SkewSlope

The skew slope is measured from the term structure at 1 month and 1 year to expiration. It is an estimate of curve slope as a rate of change in volatility in terms of moneyness, and it is calculated by taking the difference in term structure volatility between the moneyness points -0.5 and 0.5.

COMMON PROPERTIES

Primary and Secondary Axes

The Primary Axis is scaled to cover a common range for all timeseries assigned to it. This is useful for comparing similar timeseries types, such as realized and implied volatilities. There can be more than one Secondary Axis, each one associated with a single timeseries and scaled to cover the range of values over the entire period for that timeseries

Chart background

Choose Light for a white background and Dark for a gray background

Histogram buckets

User can configure the number of buckets to use for the histogram; any value between 5 and 53.

Symbol link

Automatically be updated in the Symbol 1 entry, and similarly, any time the Symbol 1 entry is changed, the selected symbol in the Symbol Viewer will be updated to the new symbol.